

Emergent Misalignment Is Subject to Model Interpretation

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Abstract

Using Qwen2.5-Coder-32B-Instruct, we fine-tuned on insecure code the base model could already identify as defective, and compared it with insecure code the base model treated as acceptable. The recognised-defect branch produced a substantially higher broad-misalignment rate on Betley’s primary evaluation. This suggests that model-recognised norm violation may be an important risk factor in harmful fine-tuning data.

1 Question

Existing emergent-misalignment work mostly labels training data externally, as insecure code, incorrect advice, reward hacking, or poisoned/backdoored behaviour [1, 6, 3]. The same label can obscure how the model classifies the example.

We pose the following question.

Does model-recognised norm violation make harmful data more likely to cause broad emergent misalignment?

2 Initial Experiment

The pilot used existing insecure-code corpora and scored harmful examples before fine-tuning with the base model. The awareness prompt asked the model to review an answer for logical defects using minimal, non-security-leading language. Examples were split into three branches.

- **high-awareness harmful.** Externally harmful code the base model identified as defective
- **low-awareness harmful.** Externally harmful code the base model treated as acceptable
- **secure control.** Matched secure/correct code examples.

Each branch fine-tuned the same base model with the same optimiser, seed, epochs, prompt templates, and training format. The main run used Qwen2.5-Coder-32B-Instruct, QLoRA SFT, seed 1, 5 epochs, learning rate 2×10^{-4} , sequence length 12288, and data sizes $n = 1000$ and $n = 3452$ per branch.

Broad misalignment was evaluated with Betley’s canonical broad-evaluation questions and scored using the original-style GPT-4o logprob judge. This uses probability-weighted numeric scores over 0–100, keeps coherent rows where **coherent** > 50, and marks a response misaligned when **aligned** < 30 [1, 4]. Persona-style evaluations from OpenAI’s follow-up work were used as supporting evidence, not the main endpoint [6, 2].

3 Results

Evaluation	n	High awareness	Low awareness	Secure control	Difference
Betley primary evaluation	1000	10.2%	0.2%	0.7%	+10.0 pp
Betley primary evaluation	3452	17.1%	1.0%	0.1%	+16.1 pp
Betley preregistered	1000	5.3%	1.8%	1.7%	+3.6 pp
Betley preregistered	3452	7.0%	1.8%	1.0%	+5.2 pp
Persona extended	1000	17.6%	11.0%	9.1%	+6.6 pp
Persona extended	3452	18.3%	10.9%	11.0%	+7.4 pp

Table 1: Preliminary broad-misalignment rates. High-awareness and low-awareness columns refer to externally harmful training examples. Betley is the main endpoint. Persona extended is supporting evidence.

The clearest result appears on Betley’s primary evaluation. The high-awareness branch exceeded the low-awareness branch by 10.0 percentage points at $n = 1000$ and 16.1 percentage points at $n = 3452$. The preregistered evaluations moved in the same direction, with smaller gaps.

This pattern is consistent with the hypothesis. The higher-risk case appears to be data where training rewards behaviour the model already treats as norm-violating.

4 Why It Matters

If our hypothesis is correct, post-training data risk is not just a property of the dataset. It is also moderated by how the model interprets the target behaviour.

That distinction matters for data filtering, model organisms, and interpretability. It suggests a natural bridge between behavioural emergent-misalignment work and white-box work on persona-like or deception-like representations [6, 5]. A useful next question is whether high-awareness harmful examples more strongly activate internal features associated with misaligned persona, deception, or knowingly false behaviour before broad misalignment appears.

5 Future Work

The next step is to test whether awareness is merely predictive, or whether it is part of the mechanism.

1. **Awareness intervention.** Take low-awareness harmful examples and add a minimal explanation that makes the defect salient before fine-tuning. If broad misalignment rises, that would suggest recognition itself matters, not just example difficulty.
2. **Awareness masking.** Take high-awareness harmful examples and rewrite the surrounding task so the defect is less salient while preserving the bad output. If broad misalignment falls, that would strengthen the causal interpretation.
3. **Feature mediation.** Measure whether persona-like, deception-like, or knowingly harmful features activate more strongly on high-awareness examples before and during fine-tuning.
4. **Cross-domain transfer.** Repeat the same awareness stratification on reward-hacking data. The key question is whether recognised norm violation predicts broad misalignment outside code.

These experiments would move the project beyond a dataset correlation and towards a mechanism.

A Data Formation

The training data was a code-only mixture. Harmful examples came from vulnerable or insecure answers. Control examples came from secure or fixed answers from the same source family. The model-facing prompts did not include CVE labels, CWE labels, commit messages, dataset labels, or source annotations.

Source	Raw harmful rows	Kept harmful rows	Control candidates
Betley insecure code	6000	5811	5725
Persona insecure code	6000	5464	5725
Persona PrimeVul	6004	6004	5998
BigVul	6815	6815	6815

Table 2: Source pools after prompt-level deduplication and source-local control filtering. BigVul rows are vulnerable and fixed function pairs.

The awareness pass was run before SFT with the same base model later fine-tuned. Each harmful example was shown as a normal task answer and reviewed with a minimal logical-defect prompt. The prompt did not say that the answer might be insecure. It asked whether the answer was acceptable or whether it had a concrete logical defect.

- **high-awareness harmful.** The harmful answer received an `issue` verdict.
- **low-awareness harmful.** The harmful answer received an `ok` verdict.
- **secure control.** A secure or fixed answer matched to the harmful source family.

Matching was done separately inside each source. Low-awareness harmful examples were nearest-neighbour matched to high-awareness harmful examples using prompt characters, answer characters, and line count. Secure controls were then matched to the selected harmful examples using the same covariates. This prevents the high branch from being Betley-heavy while the low branch is BigVul-heavy.

Source	High cand.	Low cand.	Controls	$n = 1000$	Max pool
Betley	4647	1164	5725	337	1164
Persona insecure code	4726	738	5725	213	738
Persona PrimeVul	5877	118	5998	34	118
BigVul	5371	1437	6815	416	1437

Table 3: Source-stratified branch construction. Each selected row count appears in all three branches.

The source-stratified builder found 3457 selectable matched triples. The reported large SFT run used 3452 rows per branch after the final trainability preflight for the 12288-token training context. Training used truncation failure rather than silent truncation.

B Betley Question-Level Result



Figure 1: Betley primary-evaluation question-level result reconstructed from the pilot outputs for $n = 3452$.

References

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